

Ethanol Blending in Gasoline - Nicaragua

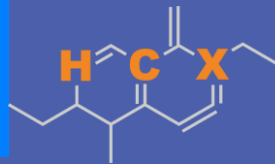
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**U.S. GRAINS &
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FARO90



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Ethanol Blending in Latin America

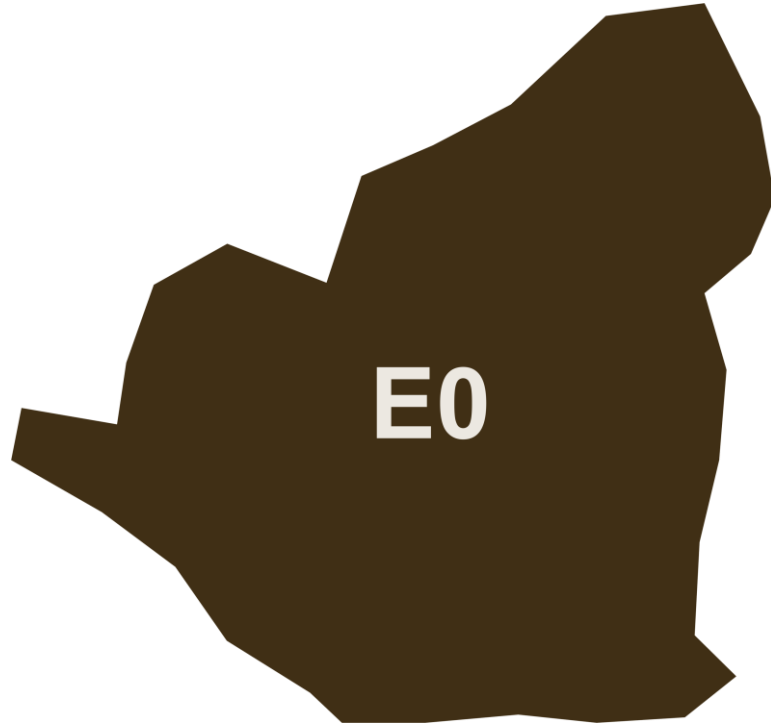
There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.



Ethanol Blending in Gasoline - Nicaragua

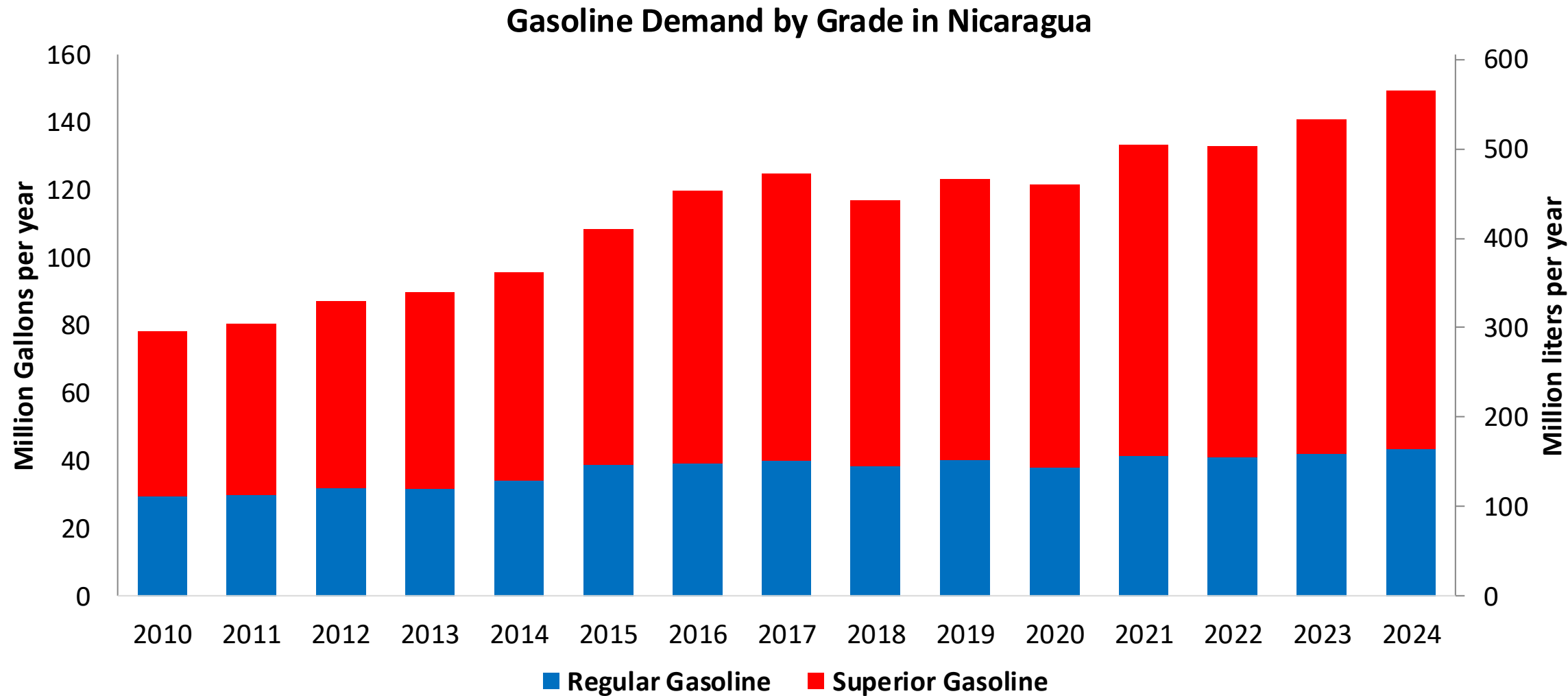


In 2024, gasoline demand was 149 million gallons (565 million liters). Nicaragua is the only country in Central America that has active refinery capacity, which produced 33% of gasoline national demand. The rest of demand is imported, previously from Venezuela and since 2016 mainly from United States. There are two gasoline grades in the country, regular with 29% and premium with 71% of market share.

According to the Ministry of Energy and RTCA, Nicaragua has no ethanol mandate.

Source: MEM Nicaragua, RTCA

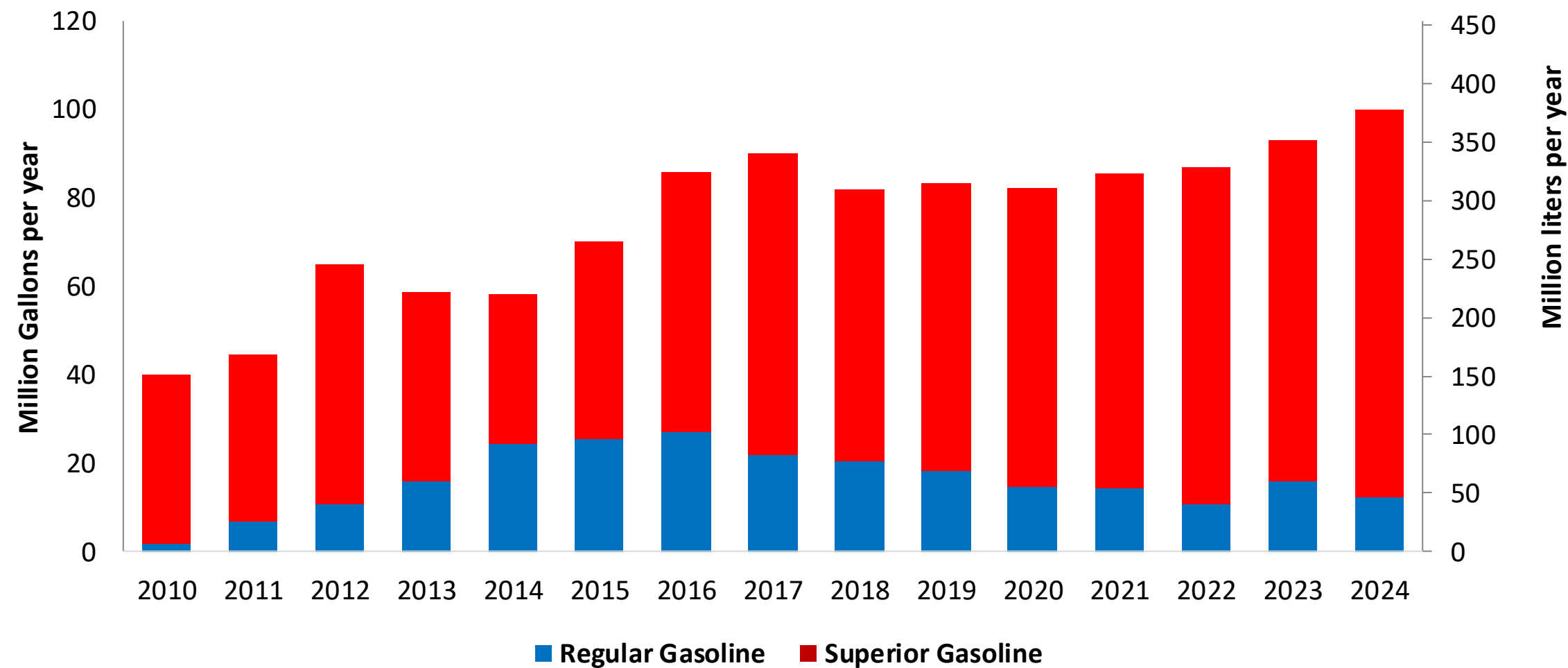
Gasoline Demand in Nicaragua



Source: MEM Nicaragua

Gasoline Imports in Nicaragua

Gasoline Imports by Grade to Nicaragua



Source: MEM Nicaragua

The FARO90 logo is shown in a blue box, followed by a chemical structure of a substituted cyclohexadiene. The structure features a six-membered ring with two double bonds. Substituents include a hydrogen atom (H), a carbon atom (C), and a group labeled 'X' (which is a vinyl group, -CH=CH₂). There are also methyl groups attached to the ring.

This stacked bar chart displays the annual consumption of Regular Gasoline (blue) and Superior Gasoline (red) from 2010 to 2024. The left Y-axis measures consumption in Million Gallons per year (0 to 60), and the right Y-axis measures it in Million liters per year (0 to 200). The x-axis lists the years from 2010 to 2024. The legend indicates that blue represents Regular Gasoline and red represents Superior Gasoline.

Year	Regular Gasoline (Million Gallons)	Superior Gasoline (Million Gallons)	Total (Million Gallons)
2010	27.5	10.5	38.0
2011	23.0	13.0	36.0
2012	21.0	1.0	22.0
2013	20.0	15.0	35.0
2014	16.0	18.0	34.0
2015	21.5	14.5	36.0
2016	18.5	14.0	32.5
2017	24.5	11.5	36.0
2018	23.5	11.0	34.5
2019	22.0	17.0	39.0
2020	23.5	16.5	40.0
2021	27.0	20.5	47.5
2022	30.5	15.5	46.0
2023	26.0	21.5	47.5
2024	31.5	18.0	49.5

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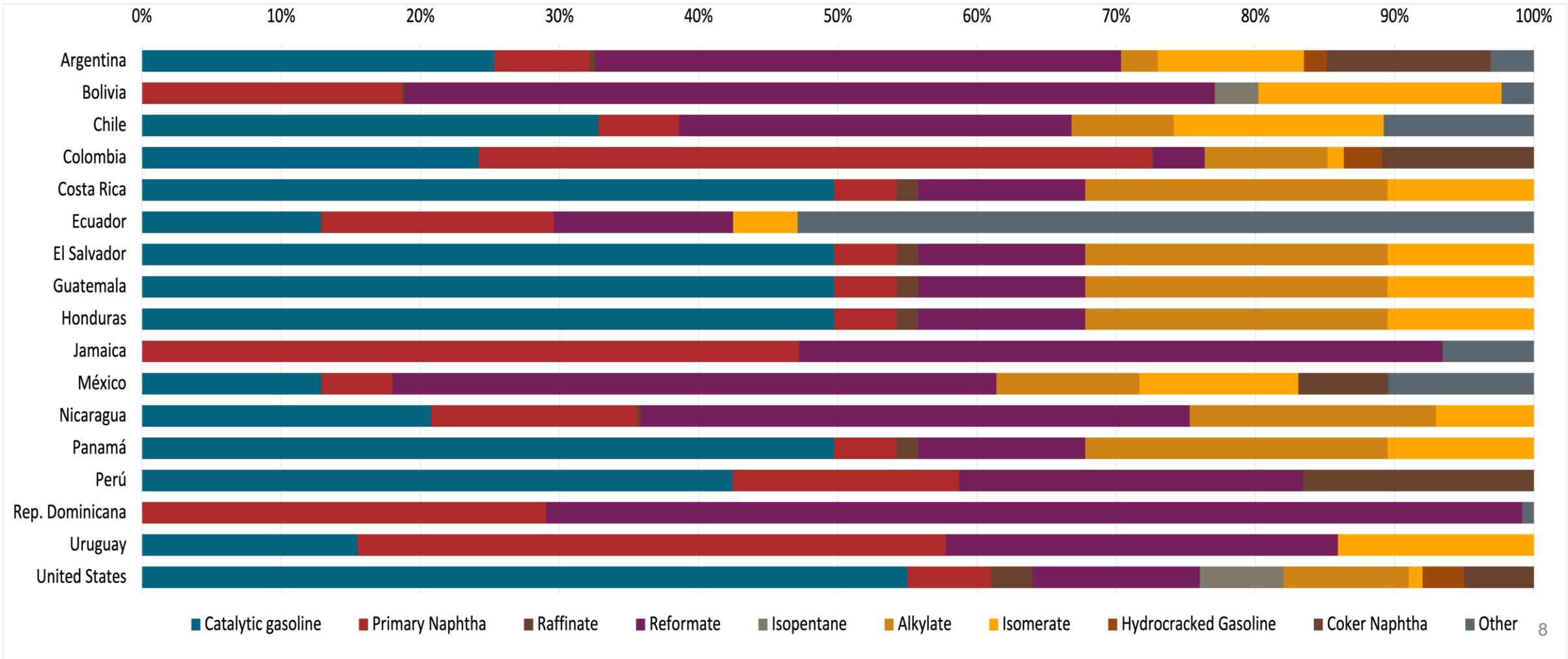
Gasoline Quality in Nicaragua

Name	RTCA 75.01.20:19	RTCA 75.01.19:19	EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2021	2021	2017			
Applicability	Whole country	Whole country	All countries			
Selected Grade	Gasoline Premium	Gasoline Regular	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	5% v/v	5% v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	50% v/v	50% v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	30% v/v	30% v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	Report	Report	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 95	> 91	> 95	> 95	> 98	> 98
MON	-	-	> 85	> 88	> 85	> 88
AKI						
Sulfur Content	< 500 mg/kg	< 500 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	2,7% v/v	2,7% v/v	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	-	-	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 69 kPa	< 69 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)						
RVP 37.8°C (Transition)						
MTBE	-	-	-	-	-	-
Ethers 5 or more C Atoms	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Source: RTCA

Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



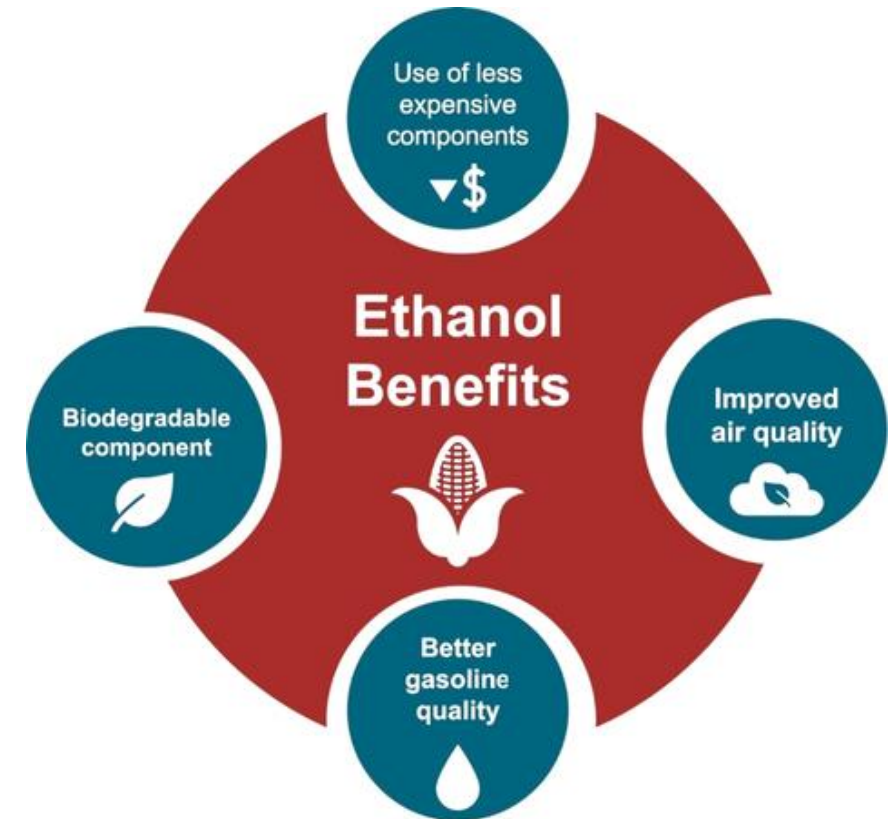
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

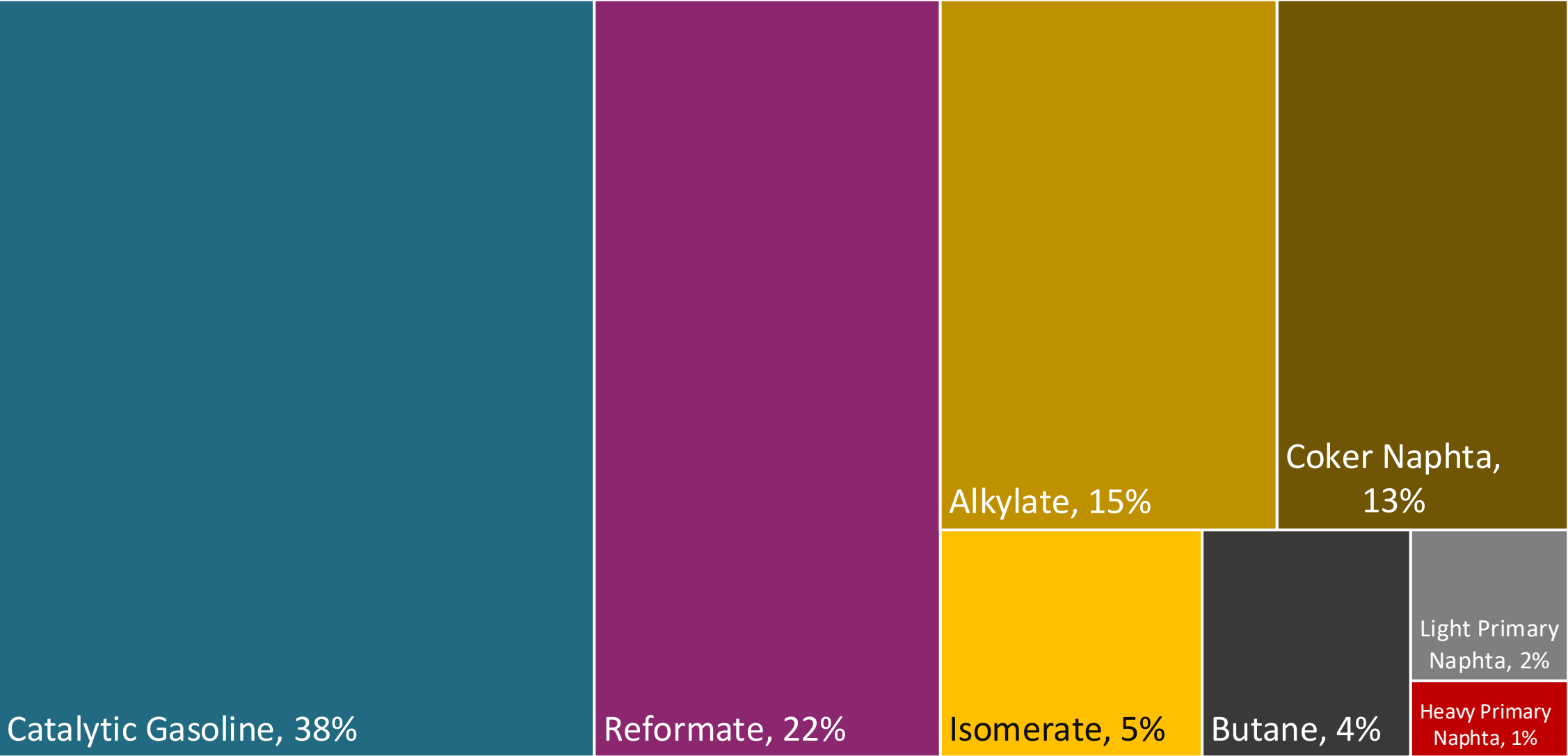
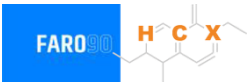
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

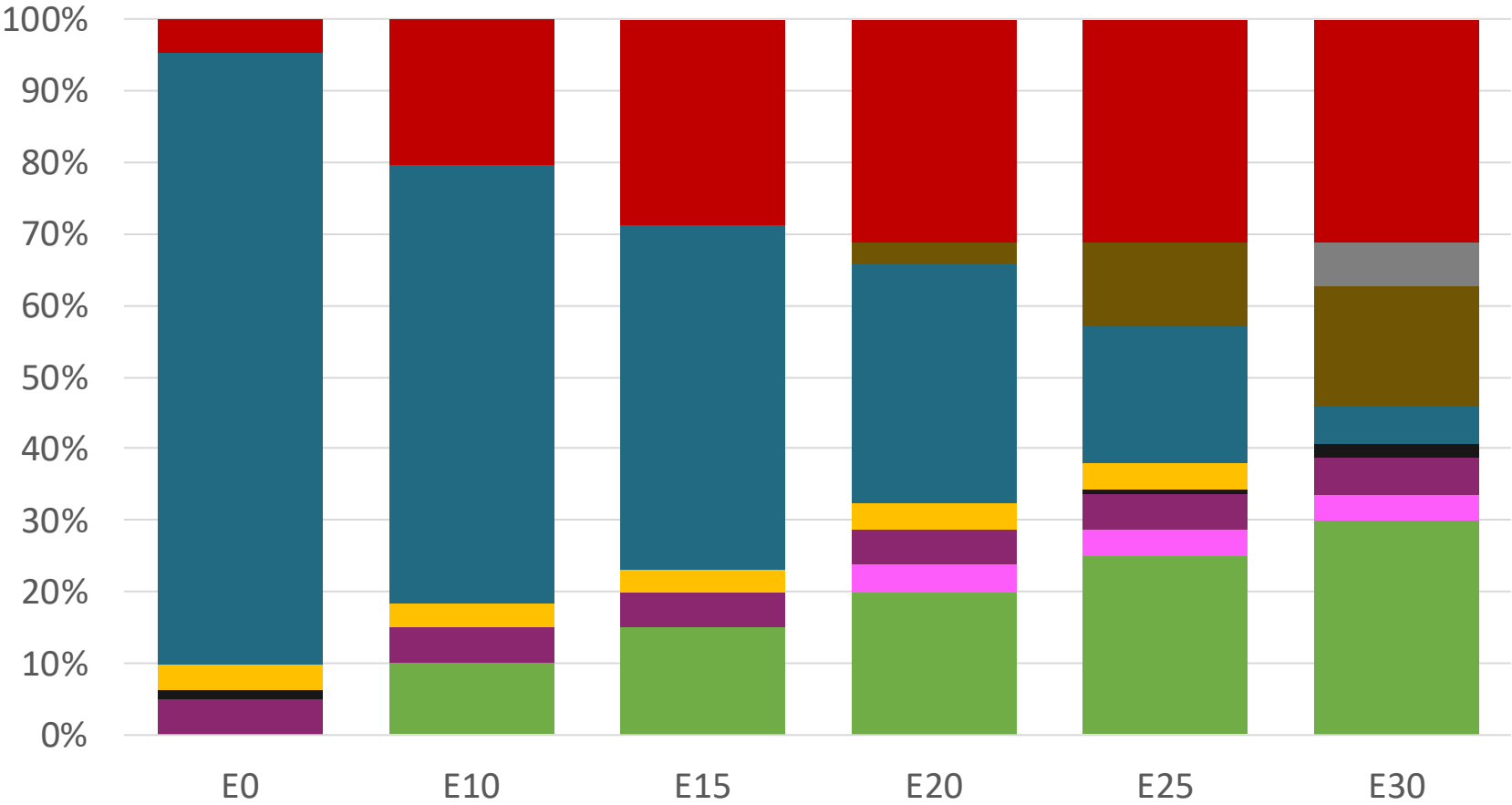
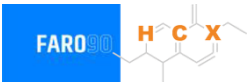


Blending Components - Nicaragua



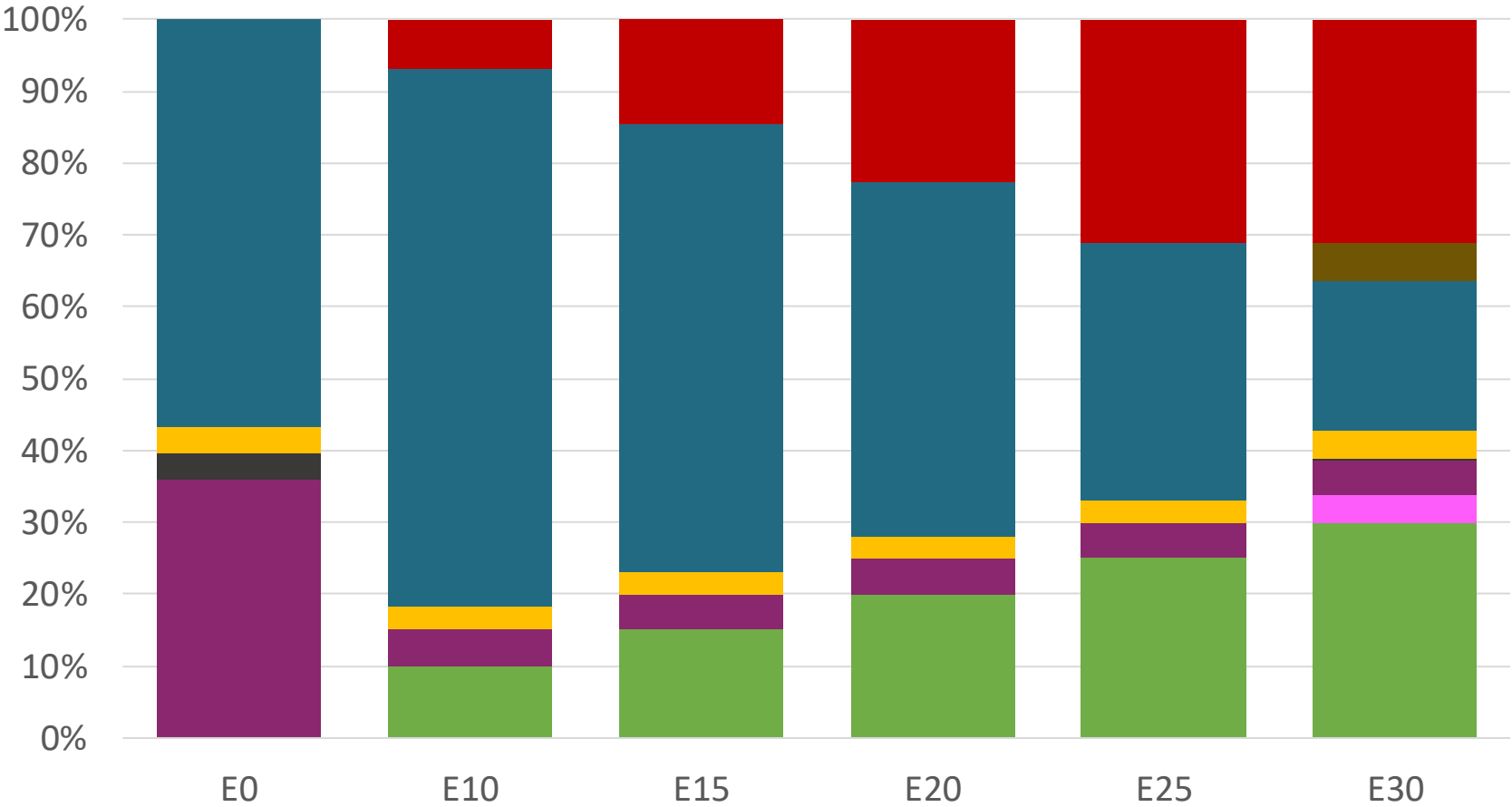
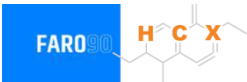
Source: Faro90

Nicaragua – Regular – Constant Octane



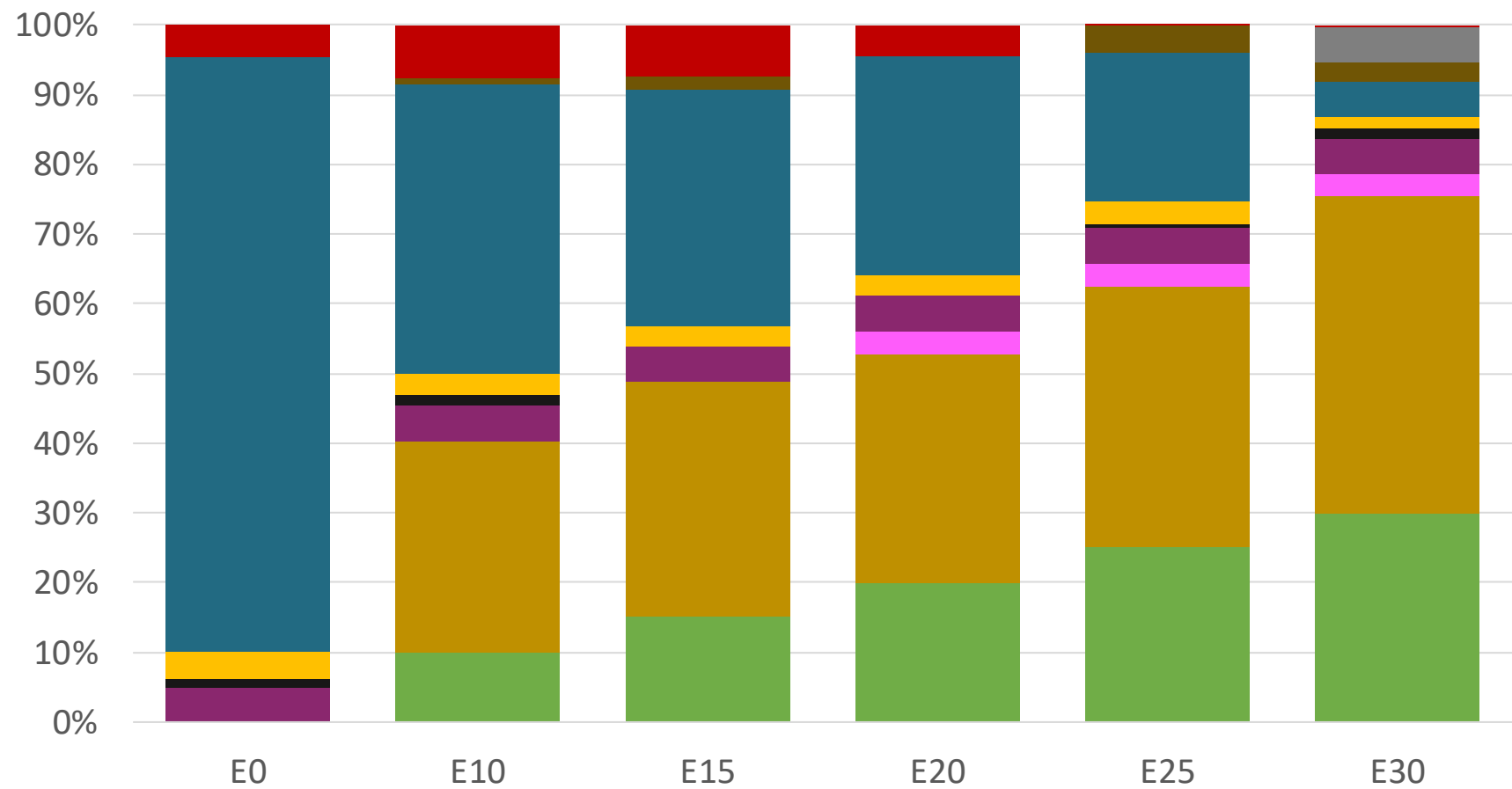
Octane (RON)	91.0	91.0	91.0	91.0	91.0	91.0
Price (USD/gal)	\$ 2.156	\$ 1.885	\$ 1.734	\$ 1.592	\$ 1.447	\$ 1.477

Nicaragua – Premium – Constant Octane



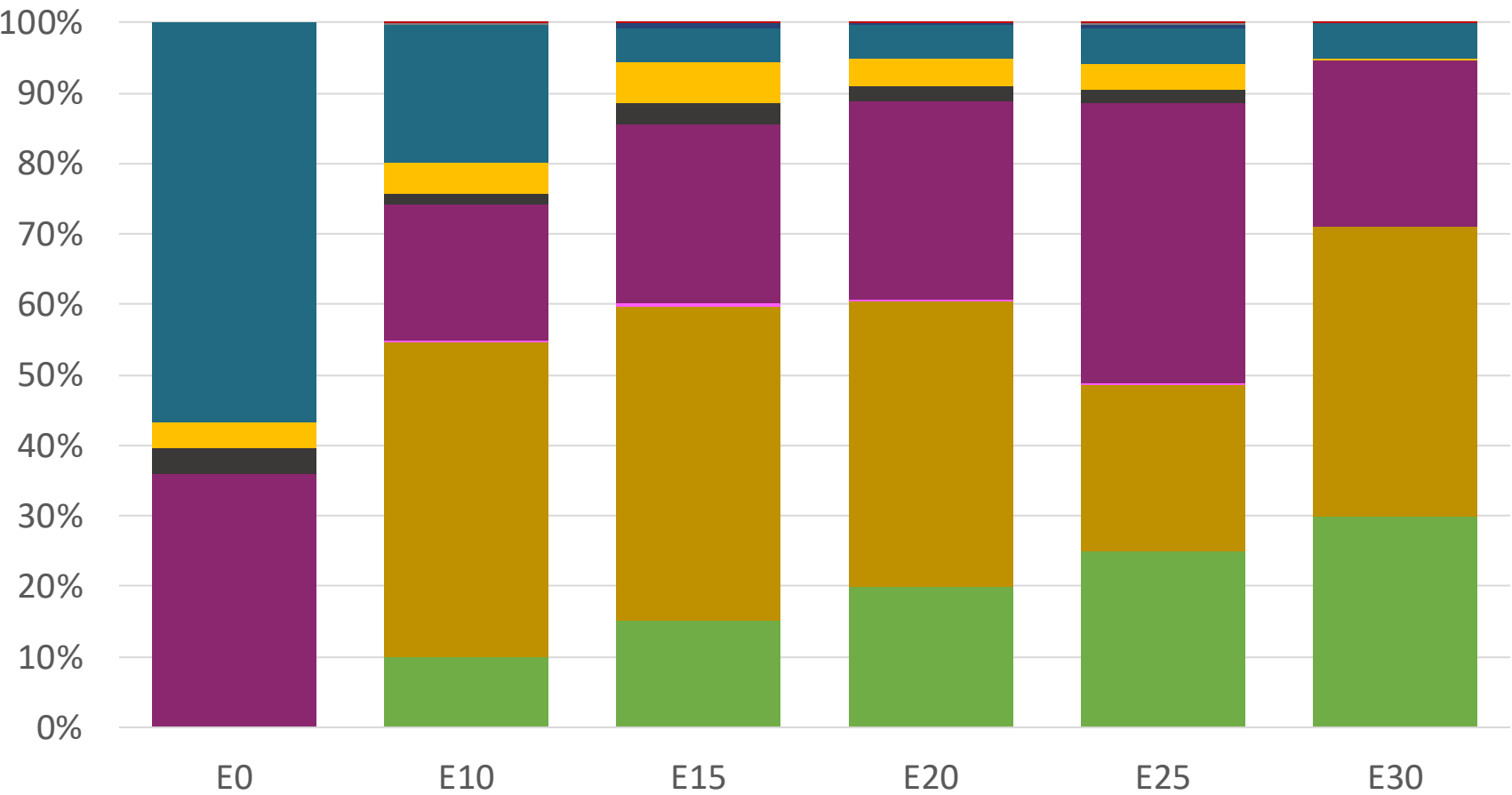
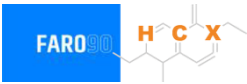
Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (USD/gal)	\$ 2.536	\$ 2.233	\$ 2.099	\$ 1.957	\$ 1.807	\$ 1.659

Nicaragua – Regular – Increased Octane



Octane (RON)	91.0	94.6	96.6	99.1	101.2	102.4
Price (USD/gal)	\$ 2.156	\$ 2.156	\$ 2.156	\$ 2.156	\$ 2.156	\$ 2.156

Nicaragua – Premium – Increased Octane



Octane (RON)	95.0	98.0	100.3	102.6	104.9	105.8
Price (USD/gal)	\$ 2.536	\$ 2.536	\$ 2.536	\$ 2.536	\$ 2.536	\$ 2.536

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

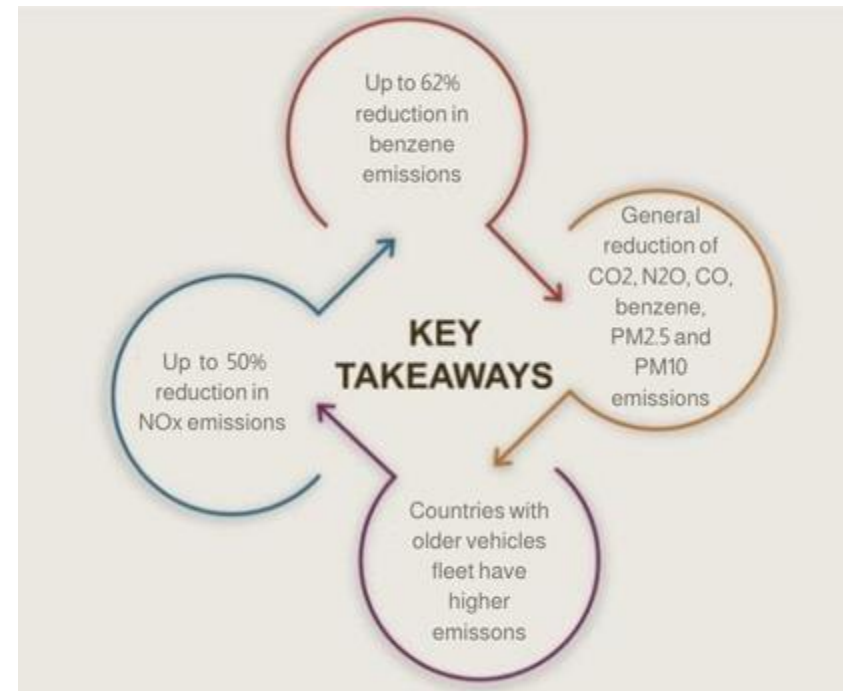
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

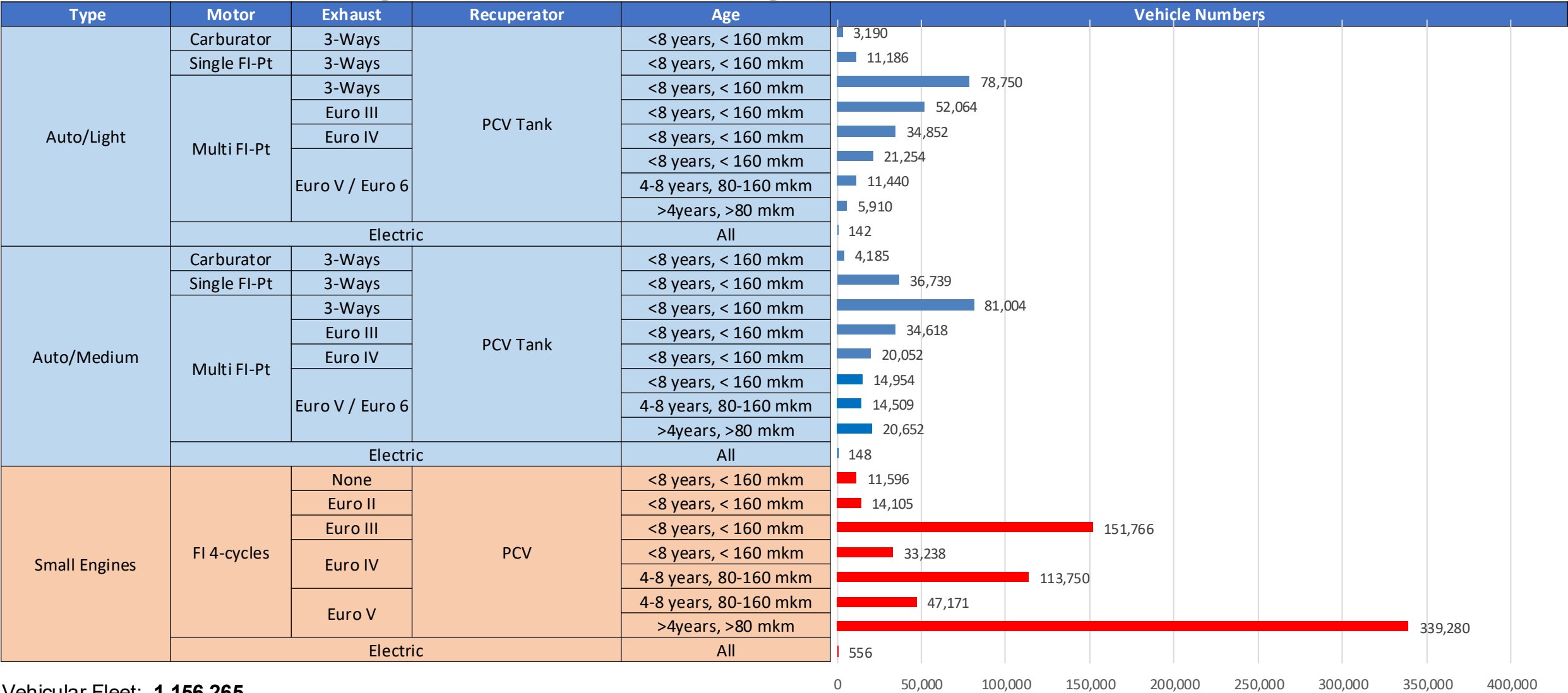
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

*Source: Bureau of transportation statistics.

Main Results



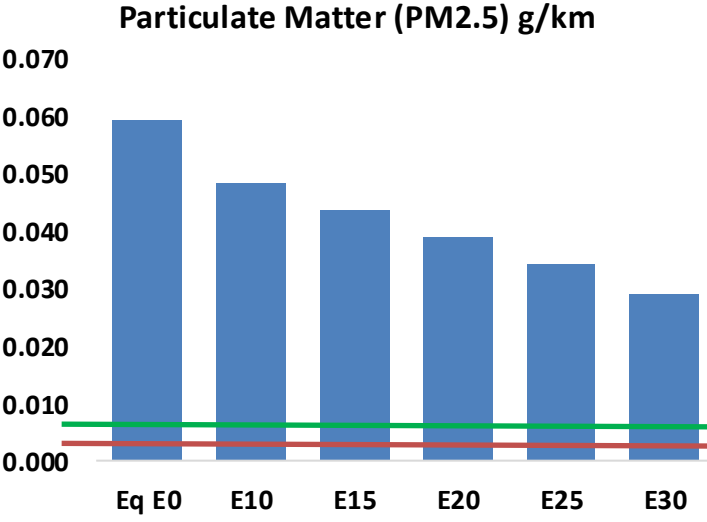
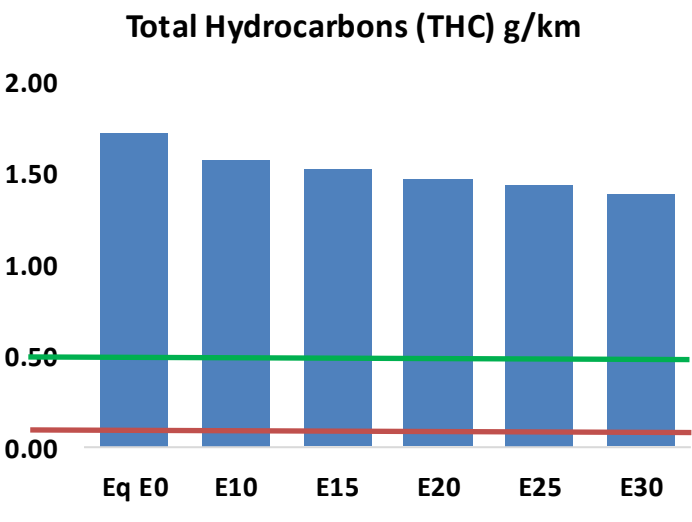
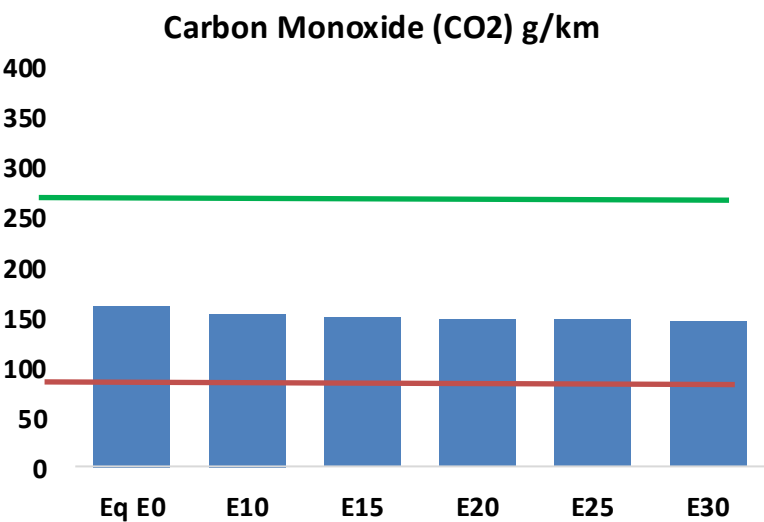
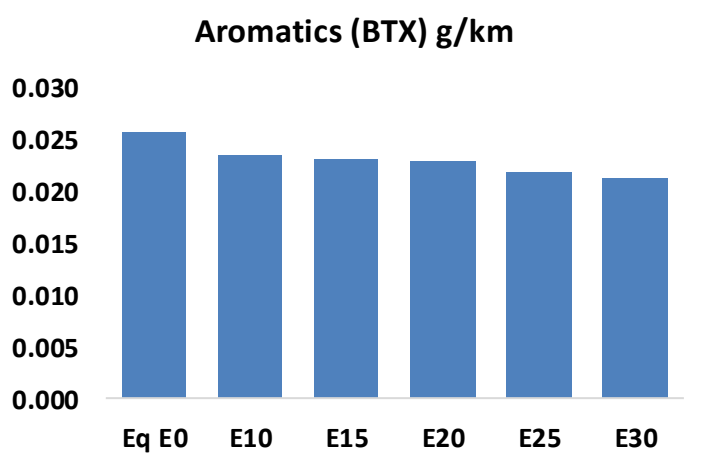
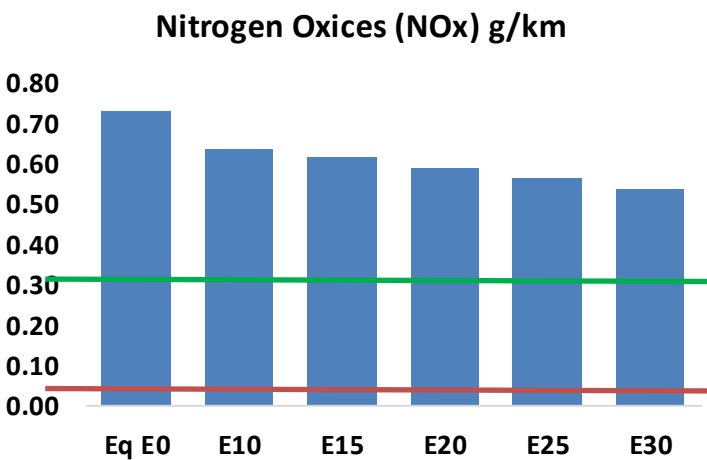
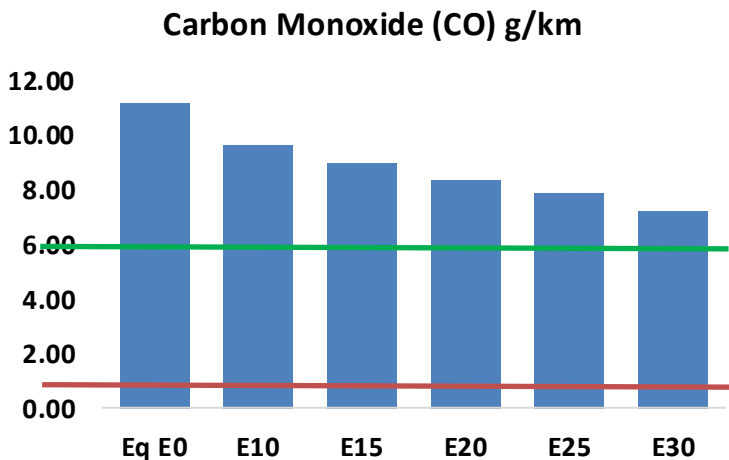
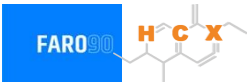
Vehicular Fleet a gasolina en Nicaragua



Vehicular Fleet: **1,156,265**
 Average Age: **11 years**
 Motorcycles: **56%**

Nicaragua - Vehicular Emissions

United States
Euro 6



Nicaragua - Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	214,932	199,644	184,355	172,068	159,814	150,425	139,047	-14%	-26%
CO2	3,079,570	3,000,381	2,925,592	2,866,772	2,837,666	2,831,064	2,778,880	-5%	-8%
VOC	33,116	31,642	30,168	29,170	28,261	27,591	26,595	-9%	-15%
NOx	14,004	12,931	12,209	11,848	11,259	10,802	10,260	-13%	-20%
SOx	35	33	30	28	26	23	21	-15%	-28%
NH3	1,316	1,303	1,290	1,296	1,311	1,321	1,329	-2%	0%
N2O	52	52	52	52	53	54	54	-1%	2%
CH4	7,490	7,490	7,490	7,602	7,767	7,909	8,044	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	148	138	128	121	113	107	99	-13%	-24%
Acetaldehyde	395	454	665	1,013	1,381	1,578	1,864	68%	249%
Formaldehyde	1,528	1,486	1,732	2,034	2,131	2,357	2,566	13%	39%
Benzene	494	475	450	443	439	420	406	-9%	-11%
PM2.5	191	173	156	141	126	110	93	-18%	-34%
PM10	945	859	773	697	623	544	461	-18%	-34%

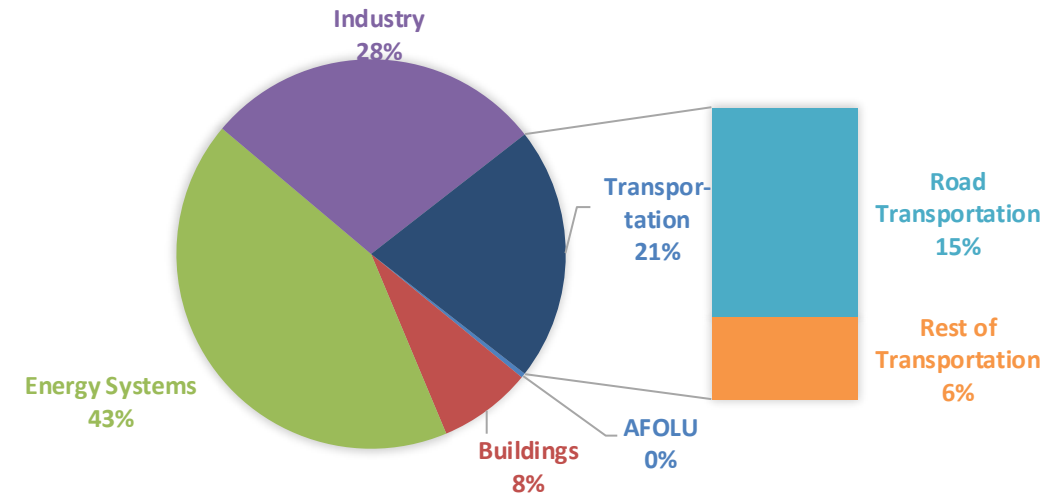
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

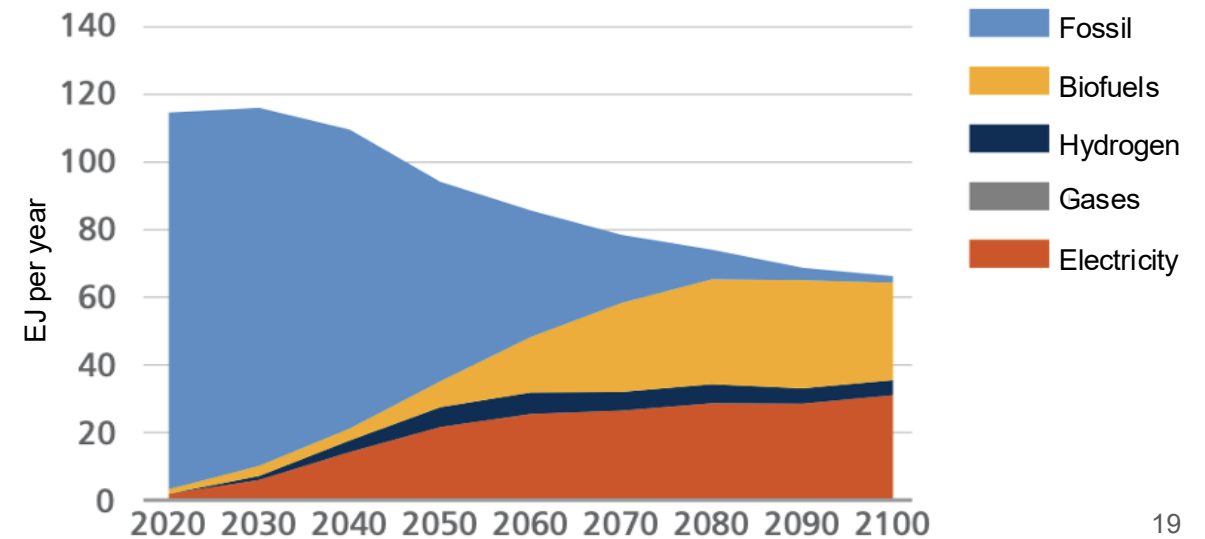
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

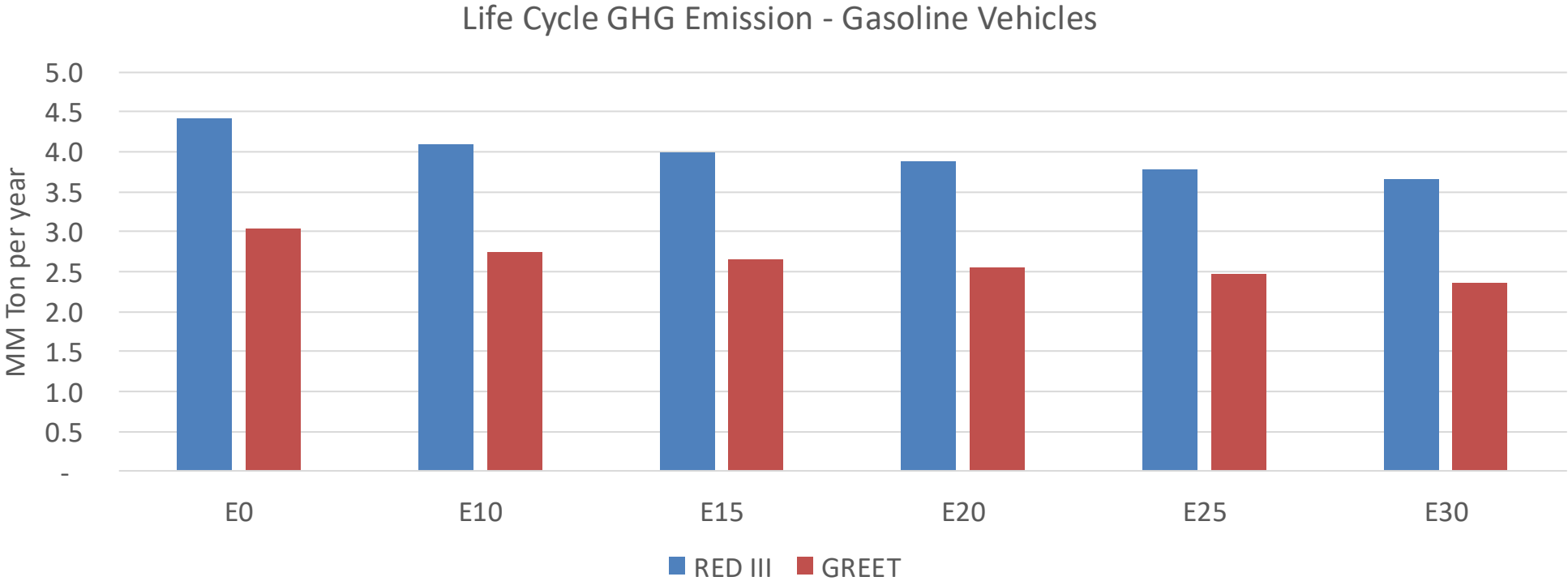
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Nicaragua – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2024 (MMT/year)	41.1
2035 Target (MMT/year)	28.8
Delta	12.3

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	10.0%	12.9%	16.3%	18.9%	22.5%
RED II/III	0.0%	7.5%	9.8%	12.4%	14.5%	17.1%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	2.5%	3.2%	4.0%	4.7%	5.5%
RED II/III	0.0%	2.7%	3.5%	4.4%	5.2%	6.1%